Elenous

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The team proposes the development of a context-aware support application designed for individuals with early-stage dementia. The application aims to help users not only remember individual tasks, but also regain their orientation in everyday life by understanding their current context. It features a highly simplified patient interface, routine support and caregiver integration for monitoring and notifications.

1. Requirements

1.1 Functional requirements

<FR1>: The system must allow the users to log in with email and password.
Description: The system allows users to securely log in using their email and password. This ensures that only authorized users can access the application and their personal data. It also helps connect the patient with the caregiver.
User priority: 4/5
Technical priority: 3/5

<FR2>: The system must allow the users to select a role after logging in.
Description: After signing in, the system allows users to select their role (patient or caregiver). This ensures that the appropriate interface and functionalities are displayed based on the user's role.
User priority: 4/5
Technical priority: 3/5

<FR3>: The system must allow the caregiver to match with their patient
Description: After selecting roles, the caregiver should connect with their patient, to manage their daily activity program, their medication list, their contact catalog and other related features.
User priority: 5/5
Technical priority: 4/5

<FR4>: The system must allow the caregiver to manage the daily activity program for the patient.
Description: The caregiver can add and edit the patient's daily activities. It provides structure and supports patients who have memory difficulties.
User priority: 5/5
Technical priority: 4/5

<FR5>: The patient must be able to view the current activity.
Description: The patient can view the activity they are supposed to perform at the current

time. It helps the patient stay oriented and follow their daily routine.
User priority: 5/5
Technical priority: 4/5

<FR6>: The system must send a notification to the patient about the current activity.
Description: The system sends a notification to remind the patient about the current scheduled activity. This helps ensure that the patient does not forget important daily tasks.
User priority: 5/5
Technical priority: 4/5

<FR7>: The patient must be able to press the emergency button.
Description: The patient can press the emergency button when immediate help is needed. It ensures quick communication in urgent situations.
User priority: 5/5
Technical priority: 4/5

<FR8>: The patient must be able to confirm that they pressed the emergency button on purpose.
Description: The patient might press the emergency button by accident. Therefore they should confirm they needed help genuinely.
User priority: 5/5
Technical priority: 4/5

<FR9>: The system must notify the caregiver when the patient presses the emergency button.
Description: When the emergency button is activated, the system automatically notifies the caregiver. This allows the caregiver to respond quickly to the patient's needs.
User priority: 5/5
Technical priority: 4/5

<FR10>: The system must allow the caregiver to manage the patient's medication.
Description: The caregiver can add and edit the patient's medication treatment plan by adding the name of medication and the exact intake time. It ensures that medication information remains accurate and up to date.
User priority: 5/5
Technical priority: 4/5

<FR11>: The system must send a notification to the patient to take their medication at a specific time.
Description: The system sends a notification to remind the patient to take their medication at a specific time. This helps improve medication adherence and patient safety.
User priority: 5/5
Technical priority: 5/5

<FR12>: The system must allow the patient to confirm that they took their medication.
Description: The patient is able to confirm that they have taken their medication. It helps track medication compliance within the system.
User priority: 5/5
Technical priority: 5/5

<FR13>: The system must inform the caregiver when the patient confirms medication intake.
Description: When the patient confirms medication intake, the system informs the caregiver. This allows the caregiver to monitor whether medication has been taken as scheduled.
User priority: 5/5
Technical priority: 5/5

<FR14>: The system must inform the caregiver if the patient does not take their medication or does not respond to the confirmation request.
Description: If the patient does not confirm medication intake, the system notifies the caregiver. This allows the caregiver to intervene if the patient may have forgotten to take their medication.
User priority: 5/5
Technical priority: 5/5

<FR15>: The system should send reminders to the patient for water intake and light exercise three times per day.
Description: The system sends reminders at predefined times: 12:00, 18:00, and 21:00. These reminders support healthy habits and improve the patient's well-being.
User priority: 3/5
Technical priority: 2/5

<FR16>: The patient should be able to press the clock to hear the current date and time.
Description: The patient can press the clock on the homepage so that they can hear a voice message with the current date and time.
User priority: 3/5
Technical priority: 2/5

<FR17>: The system should display notifications on the device lock screen.
Description: The system displays notifications even when the device is locked. This ensures that the patient does not miss important reminders.
User priority: 4/5
Technical priority: 4/5

<FR18>: The system should allow the patient to report his daily physical and emotional condition.
Description: The system asks the patient once per day how they feel about their physical and

emotional condition. For physical condition, the patient can select from predefined options (e.g., tired, energetic, sick, in pain). For emotional condition, the patient can select from predefined options (e.g., happy, sad, anxious, confused). This helps caregivers monitor the patient's overall well-being.
User priority: 4/5
Technical priority: 3/5

<FR19>: The caregiver should be able to see the patient's daily condition report.
Description: The system sends the patient's responses about their physical and emotional condition to the caregiver. This allows the caregiver to monitor the patient's daily well-being and identify potential issues.
User priority: 4/5
Technical priority: 3/5

1.2 Non functional requirements

<NFR-1> (Capacity Requirements)
The system must support up to 10 concurrent users.
Description: The system must be able to operate properly when up to 10 users are using it at the same time. This ensures stable operation for small-scale deployment and testing.
User Priority: 3/5 It is moderately important because the application is intended for a limited number of users during initial use.
Technical Priority: 4/5 It is important because the system must remain stable under concurrent usage.

<NFR-2> (Security and Privacy Requirements)
The system must comply with GDPR requirements for the collection, storage and processing of personal data.
Description: The system must protect personal and health-related data in accordance with GDPR regulations. This ensures privacy, lawful processing and secure handling of sensitive information.
User Priority: 5/5 It is very important because users' personal and health data must be protected.
Technical Priority: 5/5 It is critical because compliance with privacy regulations is mandatory.

<NFR-3> (Portability Requirements)
The system must be available on mobile phones.
Description: The application must function properly on both smartphones and tablets so that caregivers and patients can access it from portable devices.
User Priority: 5/5 It is very important because the application is intended to be used on portable devices.
Technical Priority: 4/5 It is important because the interface must adapt to different screen

sizes.

<NFR-4> (Performance Requirements)
The system must have a response time of less than 2 seconds for all main user actions.
Description: The system must respond in less than 2 seconds when the user opens the home screen, views activities, confirms medication intake or accesses contacts. This ensures a smooth and efficient user experience.
User Priority: 4/5 It is important because slow response may confuse or frustrate elderly users.
Technical Priority: 5/5 It is important because performance directly affects usability and system quality.

<NFR-5> (Accessibility and Usability Requirements)
The system must provide large buttons to the patient's screen.
Description: The interface must use large buttons to improve readability for elderly users and users with visual or cognitive difficulties.
User Priority: 5/5 It is very important because readability and accessibility are essential for the target users.
Technical Priority: 4/5 It is important because the user interface must be designed specifically for accessibility.

1.3 Glossary

User: Any person who interacts with the system. In this application, users include the patient and the caregiver.

Patient: A person with memory difficulties or early-stage dementia who uses the application to receive reminders, view activities and confirm tasks such as medication intake.

Caregiver: A person responsible for assisting and supervising the patient. The caregiver can manage the patient’s activities, medication schedule, contacts and monitor whether tasks have been completed.

System: The dementia support application that helps patients follow their daily routine, receive reminders and communicate with caregivers.

Activity: A scheduled task that the patient must perform during the day, such as eating, exercising, taking medication or attending an appointment.

Daily Activity Program: A structured schedule created by the caregiver that organizes the patient’s daily activities and routines.

Notification: A message generated by the system to remind the patient about scheduled activities or medication intake

Medication Reminder: A notification sent by the system to remind the patient to take their medication at a specific time.

Medication Confirmation: The action performed by the patient to confirm that they have taken their medication.

Emergency Button: A feature that allows the patient to request immediate help by notifying the caregiver.

Activity Completion: The process in which the patient marks a scheduled activity as completed within the system.

Activity History A record of completed or missed activities stored in the system and available for caregivers to review.

Voice Notification: An audio reminder generated by the system to assist patients who may have difficulty reading text notifications.

Lock Screen Notification: A system notification that appears on the device screen even when the device is locked.

GDPR: The General Data Protection Regulation, a European regulation that ensures the protection of personal data and privacy of individuals.

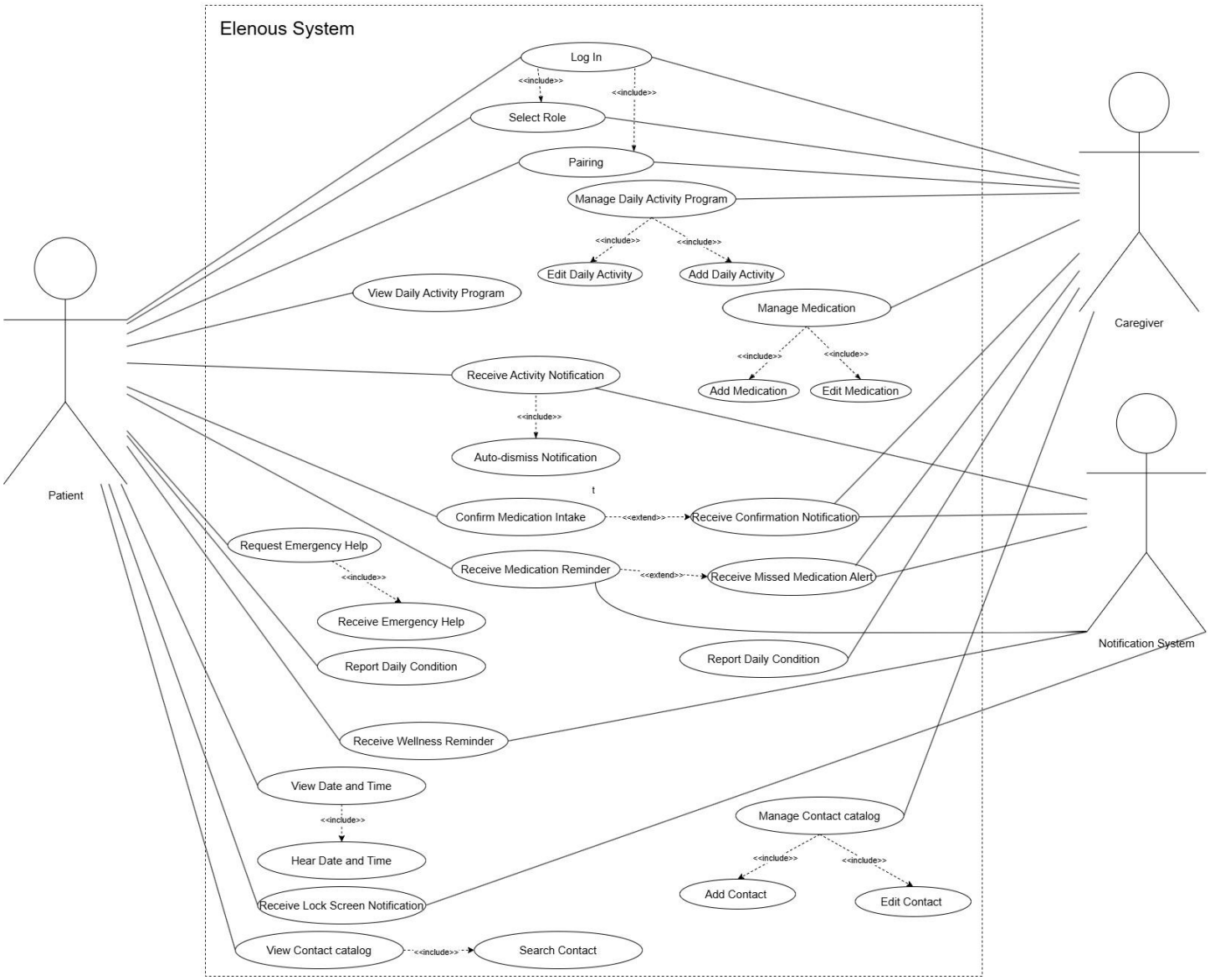
Accessibility: Design principles that ensure the application can be easily used by elderly users or users with cognitive or visual difficulties, such as large fonts and high color contrast.

Response Time: The time required for the system to respond to a user action, such as opening a screen or confirming an activity.

Concurrent Users: The number of users who can use the system simultaneously without affecting system performance.

2. Use case

2.1 Use case diagram



2.2 Activity diagrams

2.2.1 Feature 1-Log In

Log In

- As a user
- So that I can start using the app
- I want to log in and match with my caregiver/ patient

Background:
Given that I am a user
And I open the app for the first time
And I am on the login page
And email format is:
|email |[a-z, A-Z, 0-9]@gmail.com |maximum of 15|
And password format is:
| password | [A-Z, a-z, 0-9] | maximum of 10 |

Scenario: Log in with correct info

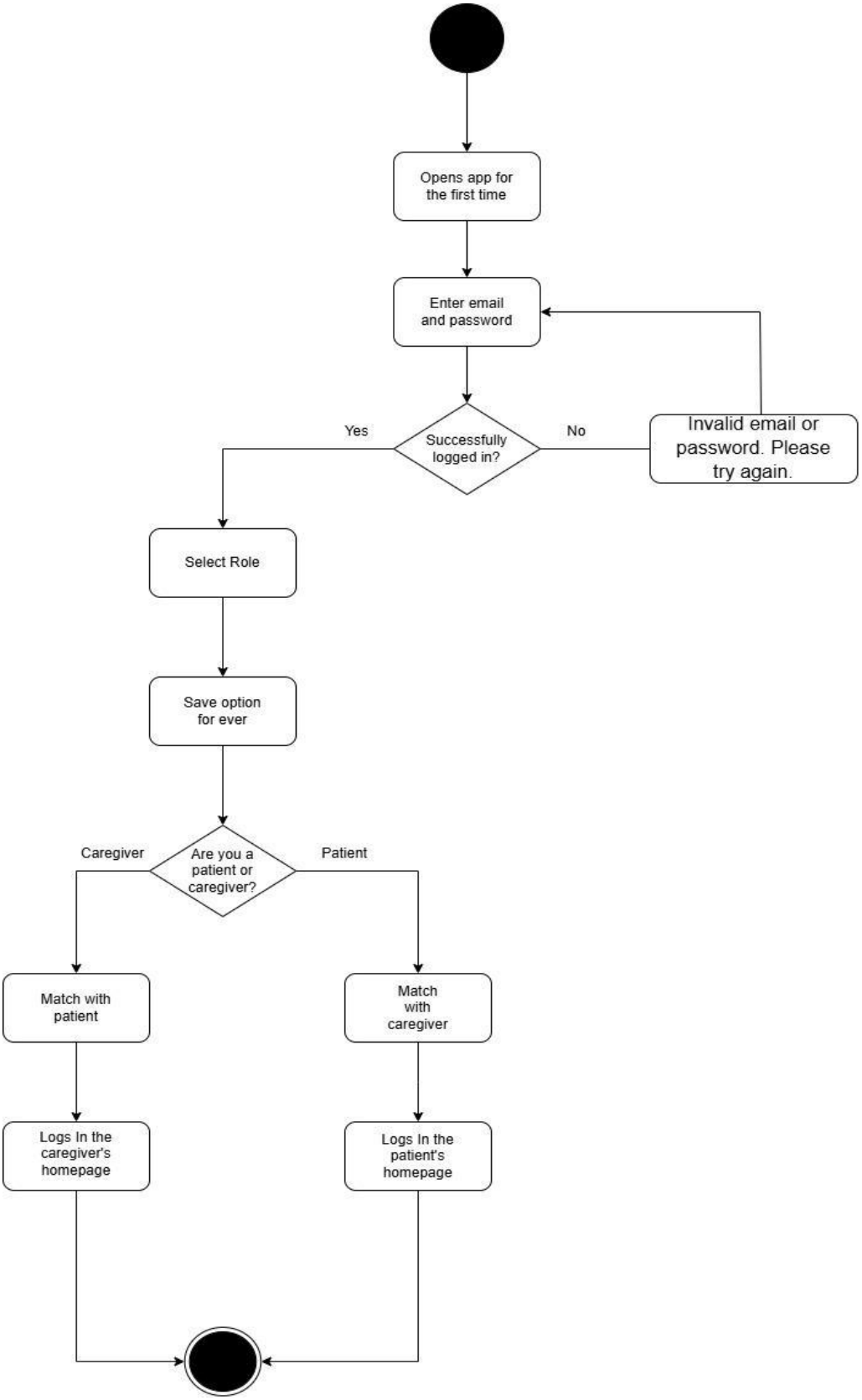
When I enter a correct email and password
And I press the log in button
Then I should see a message saying "You have logged in successfully"
And I should see the select role page

Scenario: Log in with wrong info

When I enter an incorrect email or password
And I click log in button
Then I should see a message saying "Invalid email or password. Please try again."
And I should be able to try again

Scenario: Select Role

When I have successfully logged in
And I click the button saying "Select my role"
And I should choose between the buttons "Patient" or "Caregiver"
Then I should be redirected to the appropriate page for my role
And I should be able to connect with the caregiver/ patient via email


Opens app for the first time
Enter email and password
Successfully logged in?
Yes
No
Invalid email or password. Please try again.
Select Role
Save option for ever
Are you a patient or caregiver?
Caregiver
Patient
Match with patient
Match with caregiver
Logs In the caregiver's homepage
Logs In the patient's homepage

2.2.2 Feature 2-Manage Daily Activity Program

<u>Manage Daily Activity Program</u>

- As a caregiver
- So that the patient can follow a structured daily routine
- I want to create and edit daily activities.

Background:
Given that I am the caregiver
And I am logged in
And I am on the homepage
And activity time format is:
| time | [00–23]:[00–59] |
And activity date format is:
| date | [01–31]/[01–12]/[2026–2100] |

Scenario: Add daily activity successfully

When I press the daily activity program
And I press Add New Daily Activity
Then I enter valid activity name, activity date and activity time
Then I should see a message saying "New daily activity added successfully"
And I should be able to see the new activity in the daily activity program
And I should be able to go back to homepage

Scenario: Edit daily activity successfully

When I press the daily activity program
And I press "Edit Daily Activity"
And at least one activity already exists on the program
And I select the activity I want to edit
Then I edit a valid activity name, activity date and activity time
Then I should see a message saying "Daily activity edited successfully"
And I should be able to see the edited activity in the daily activity program
And I should be able to go back to homepage

Scenario: Edit daily activity with invalid or missing information

When I press the daily activity program
And I press "Edit Daily Activity"
And at least one activity already exists on the program

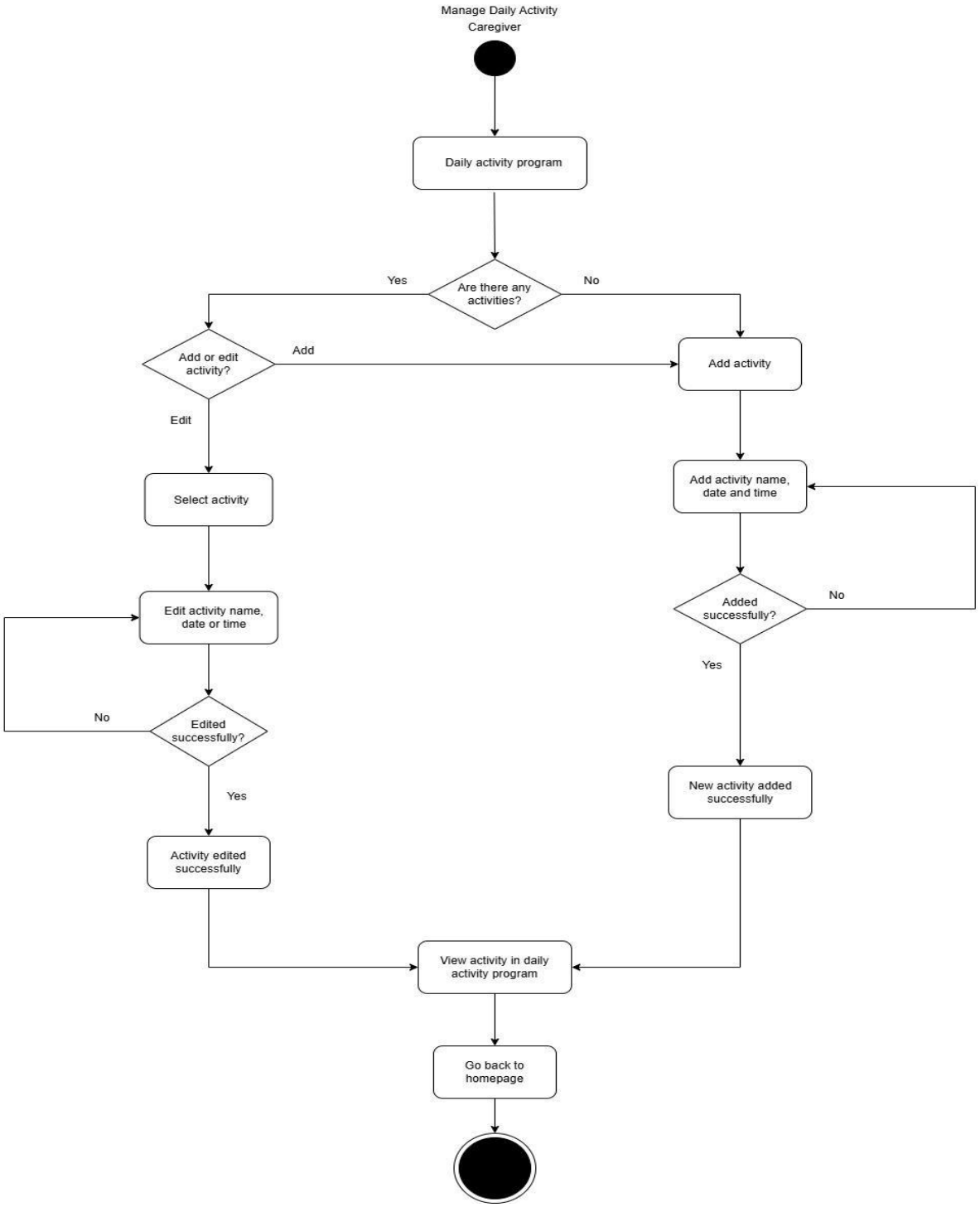
And I select the activity I want to edit

And I enter an invalid or empty activity name, activity date or activity time

Then I should see a message saying "Something went wrong! Please check for invalid or missing information!"

And I should have the option to edit the activity again

And I should have the option to return to the homepage without saving the changes



2.2.3 Feature 3-View Daily Activity Program

<u>View Daily Activity Program</u>

- As a patient
- So that I can stay oriented and know what I need to do, I want to view my daily activity program

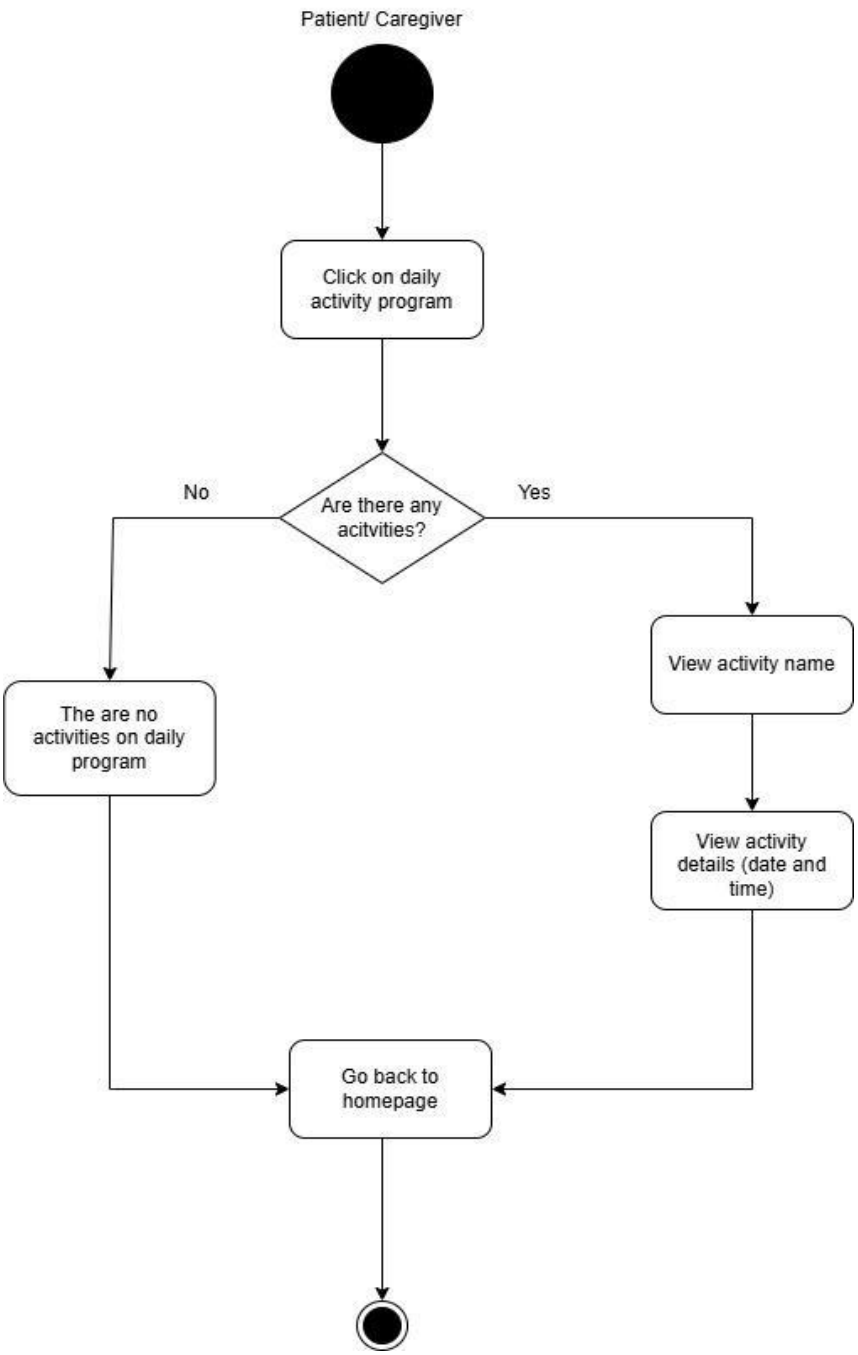
Background:
Given that I am the patient
And I am logged in
And I am on the homepage

Scenario: View current daily activity successfully

When I click "Daily Activity Program" and at least one activity exists
Then I should see the current activity, the activity date and the activity time
And I should be able to go back to homepage

Scenario: View daily activity program when no activities exist

When I click "Daily Activity Program" but no activities have been added yet
Then I should see a message saying "No daily activities available."
And I should be able to go back to homepage


Patient/ Caregiver
Click on daily activity program
Are there any acitvities?
No
Yes
The are no activities on daily program
View activity name
View activity details (date and time)
Go back to homepage

2.2.4 Feature 4-Receive Daily Activity Notification

<u>Receive Daily Activity Notification</u>

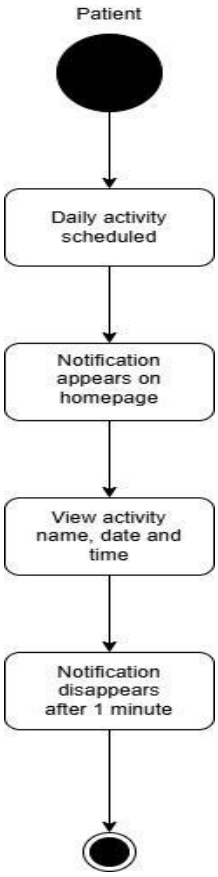
- As a patient
- So that I do not forget important daily tasks
- I want to receive notifications for scheduled activities.

Background:

Given that I am the patient
And I am logged in
And I am on the homepage
And a daily activity is scheduled

Scenario: Receive notification for daily activity

When I receive a daily activity notification
Then I should see the activity name, the activity date and the activity time
And the notification message should disappear after 1 minute



2.2.5 Feature 5- Manage Patient Medication

<u>Manage Patient Medication</u>

- As a caregiver
- So that the patient follows the correct medication schedule
- I want to add and manage medications in the patient's treatment plan.

Background:
Given that I am the caregiver
And I am logged in
And I am on the homepage
And intake time format is:
| intake time | [00–23]:[00–59] |

Scenario: Add new medication successfully

When I press the medication
And I press "Add Medication"
Then I should be asked to enter the medication name and intake time
And I enter a valid medication name and valid intake time
And I press the button save
And the system should store the medication information
And I should see the medication on medication list
And I should see a message saying "Medication added successfully."
And I should be able to go back to homepage

Scenario: Adding invalid or missing information

When I press the medication button
And I press "Add Medication"
And I enter invalid or missing information
And I press the button "Save"
Then the system should prevent saving the medication
And I should see a message saying "Please check for invalid or missing information"
And I should have the option to edit
And I should have the option to go back

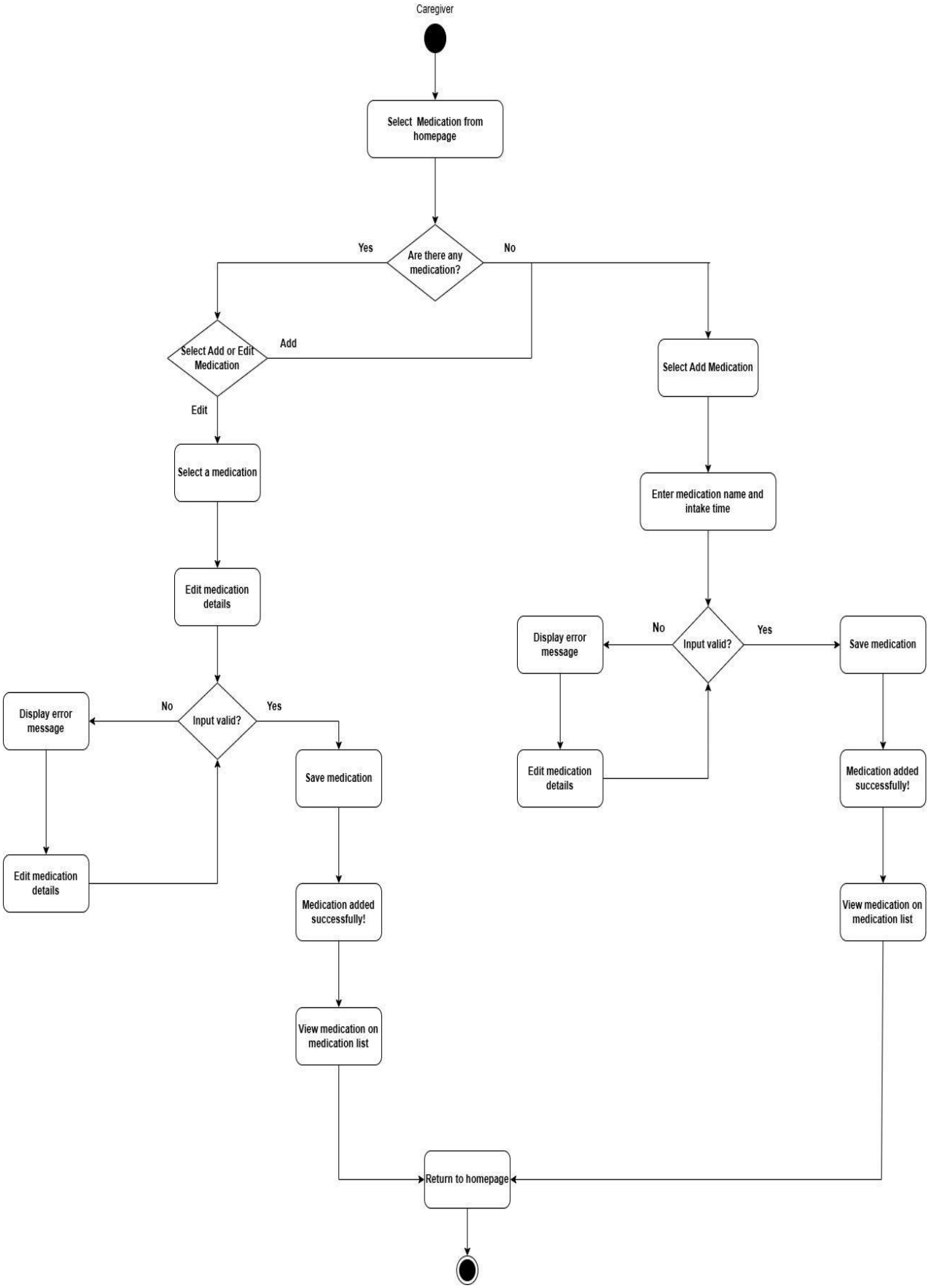
Scenario: Successfully updating medication time

When I press the medication
And I press "Edit Medication"
And I select the medication

Then I should be able to edit medication details
When I press the button "Save"
Then the system should update the medication schedule
And I should see the updated intake time in the medication list
And I should see a message saying "Medication updated successfully."
And I should be able to go back to homepage

Scenario: Invalid medication time entered

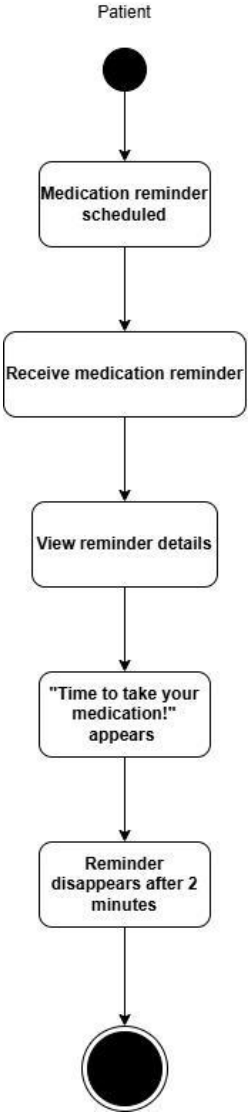
When I press the medication
And I press "Edit Medication"
And I select the medication
Then I should be able to modify the medication details
When I enter an invalid intake time
And I press the button "Save"
Then the system should reject the update
And I should see a message saying "Invalid medication time. Please enter a valid time."
And I should have the option to edit the medication time
And I should have the option to go back to homepage


Caregiver
Select Medication from homepage
Are there any medication?
Yes
No
Select Add or Edit Medication
Add
Edit
Select Add Medication
Select a medication
Enter medication name and intake time
Edit medication details
Display error message
No
Input valid?
Yes
Save medication
Display error message
No
Input valid?
Yes
Edit medication details
Save medication
Edit medication details
Medication added successfully!
Medication added successfully!
View medication on medication list
View medication on medication list
Return to homepage

2.2.6 Feature 6-Receive Medication Reminder

<u>Receive Medication Reminder</u>

- As a patient
- So that I remember to take my medication on time
- I want to receive medication reminders.



Background:
Given that I am the patient
And I am logged in
And a medication has been scheduled

Scenario: Reminder displayed at the scheduled time

When a scheduled medication time arrives
Then the system should send a medication reminder notification
And I should see the medication name on the notification
And I should see a message saying "Time to take your medication."
And the notification should disappear after 2 minutes.

2.2.7 Feature 7-Confirm Medication Intake

Confirm Medication Intake

- As a patient
- So that the system records my medication intake
- I want to confirm that I have taken my medication.

Feature: Confirm Medication Intake

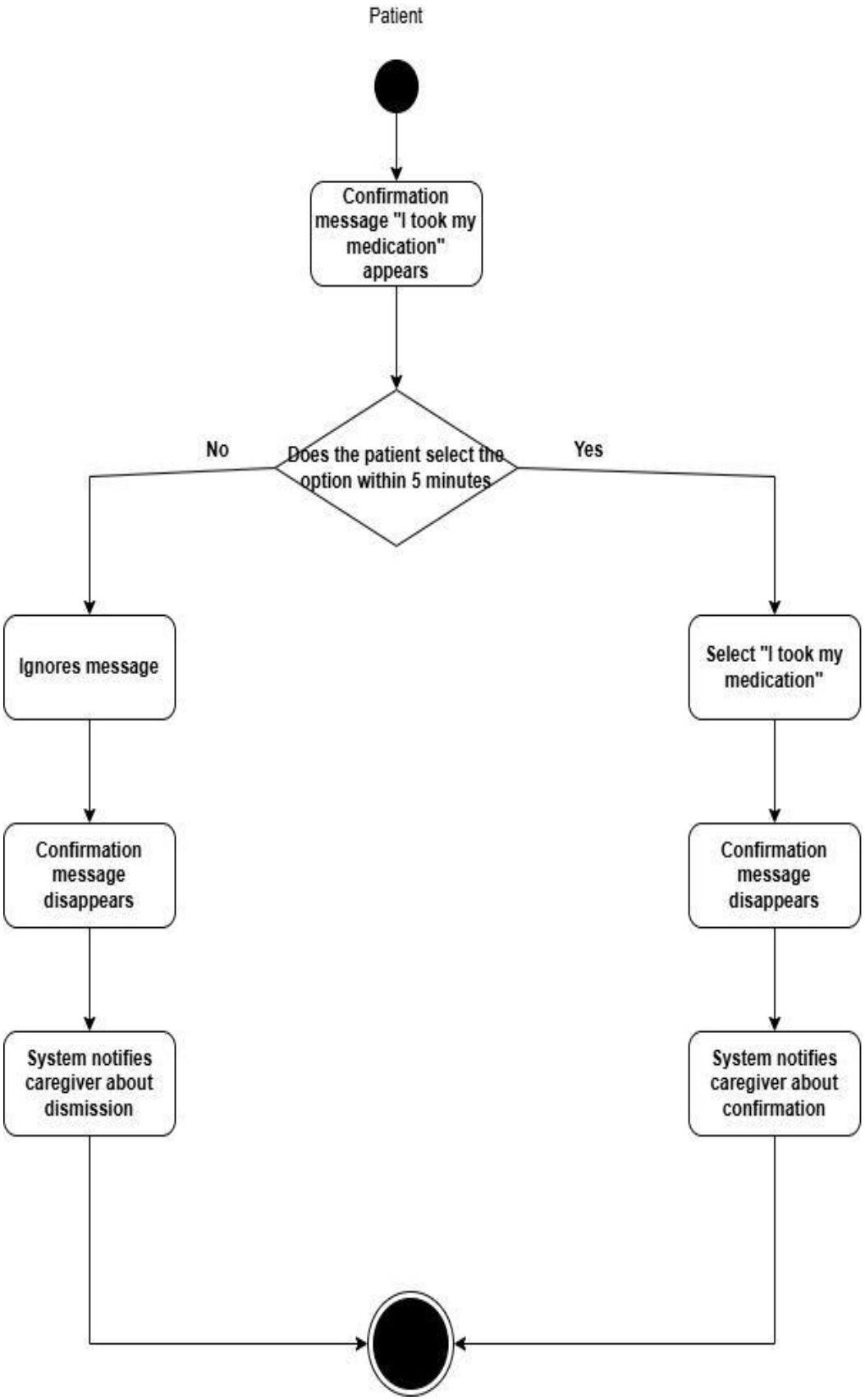
Background:
Given that I am the patient
And I am logged in
And I have received a medication reminder

Scenario: Successful medication confirmation

When 5 minutes have passed since the medication reminder came up
Then I should see the button "I took my medication"
And I should be able to press it
And it should disappear after I press it
And the system should notify the caregiver that I confirmed medication intake

Scenario: Reminder dismissed without confirmation

When 5 minutes have passed since the medication reminder came up
And the button "I took my medication" appears on screen
And I do not press it within 5 minutes
Then the reminder should disappear
And the system should notify the caregiver that I did not confirm the medication intake


Patient
Confirmation message "I took my medication" appears
Does the patient select the option within 5 minutes
No
Yes
Ignores message
Select "I took my medication"
Confirmation message disappears
Confirmation message disappears
System notifies caregiver about dismission
System notifies caregiver about confirmation

2.2.8 Feature 8-Caregiver Confirmation Notification

Caregiver Notification After Confirmation

- As a caregiver
- So that I can monitor medication adherence
- I want to be notified when the patient confirms medication intake.

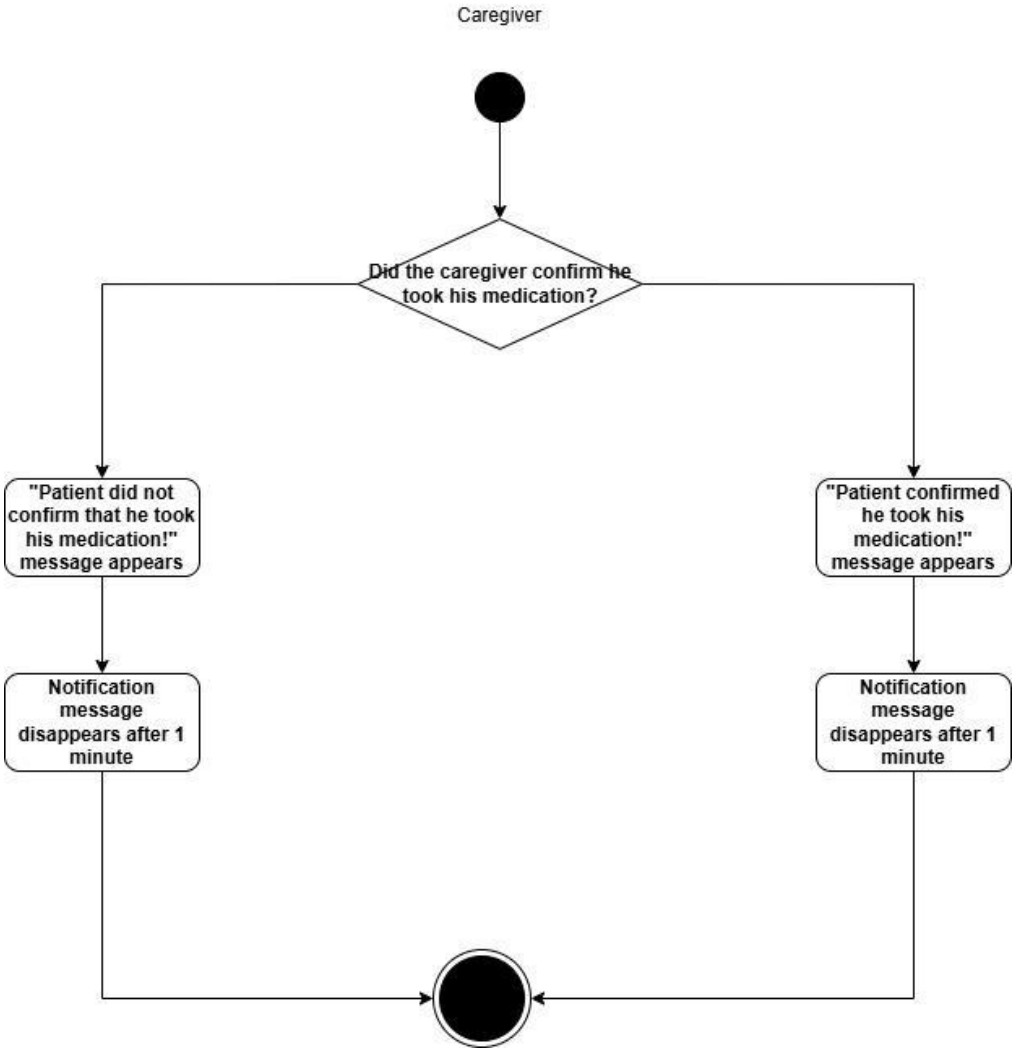
Background:
Given that the patient received the medication reminder

Scenario: Caregiver receives confirmation notification

When the patient confirms they took their medication
Then the caregiver should receive a notification saying "The patient has taken the scheduled medication."
And the notification should disappear after 1 minute

Scenario: Caregiver notified for missed medication

When the patient did not confirm they took their medication after the reminder
Then I should receive a notification message saying "The patient has not confirmed taking the scheduled medication"
And the notification should disappear after 1 minute



2.2.9 Feature 9-Press Help Button

<u>Press Help Button</u>

- As a patient
- So that I can quickly ask for help in an emergency
- I want to press the emergency/help button

Background:

Given that I am the patient

And I am logged into the system

And I am on homepage

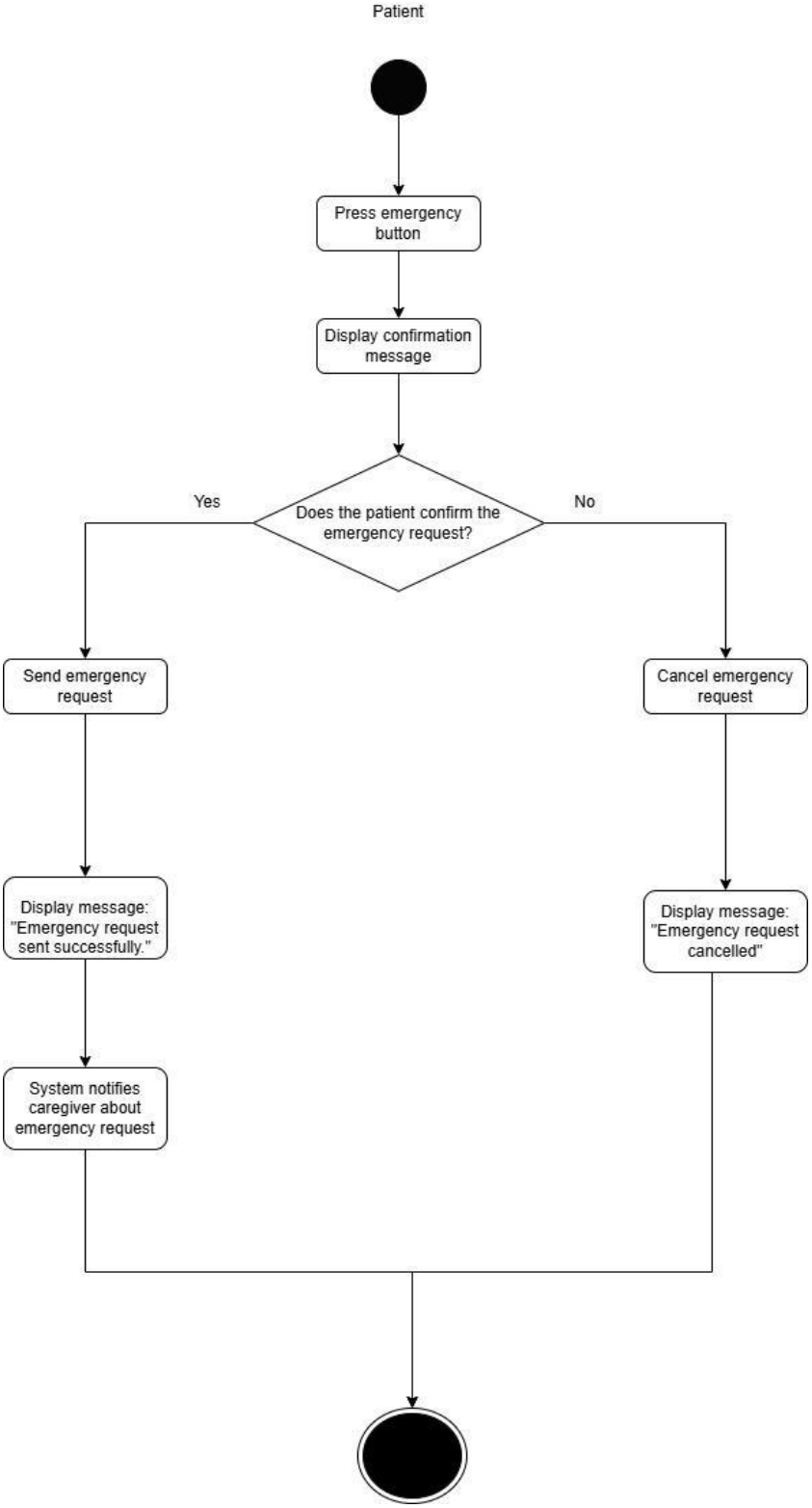
And the emergency button is visible on the screen

Scenario: Successfully pressing the emergency button

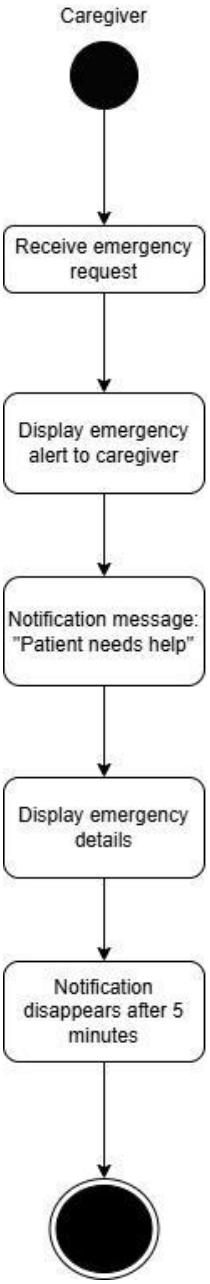
When I press the emergency button
Then the system should ask me to confirm the emergency request
And I should be able to confirm
And the system should display a confirmation message saying "Emergency request sent successfully."
And the system should notify the caregiver about the emergency

Scenario: Emergency button pressed accidentally

When I press the emergency button
Then the system should ask me to confirm the emergency request
And I should be able to cancel the request before it is sent
And the system should not notify the caregiver about the emergency


Patient
Press emergency button
Display confirmation message
Does the patient confirm the emergency request?
Yes
No
Send emergency request
Cancel emergency request
Display message: "Emergency request sent successfully."
Display message: "Emergency request cancelled"
System notifies caregiver about emergency request

2.2.10 Feature 10-Caregiver Receives Emergency Alert



<u>Caregiver Receives Emergency Alert</u>

- As a caregiver
- So that I can respond quickly to emergencies
- I want to receive an alert when the patient presses the emergency button.

Background:
Given that I am the caregiver
And I am logged into the system
And the patient presses the emergency button

Scenario: Successfully receiving emergency alert

When the system sends the emergency notification
Then I should receive an alert message saying "Patient needs help."
And I should see the patient's name and the time of the emergency request
And the notification should disappear after I press the dismiss button

2.2.11 Feature 11-Daily Physical And Emotional Condition

<u>Daily physical and emotional condition</u>

- As a patient
- So that I can inform my caregiver about my condition
- I want select some states that describe how I am feeling today

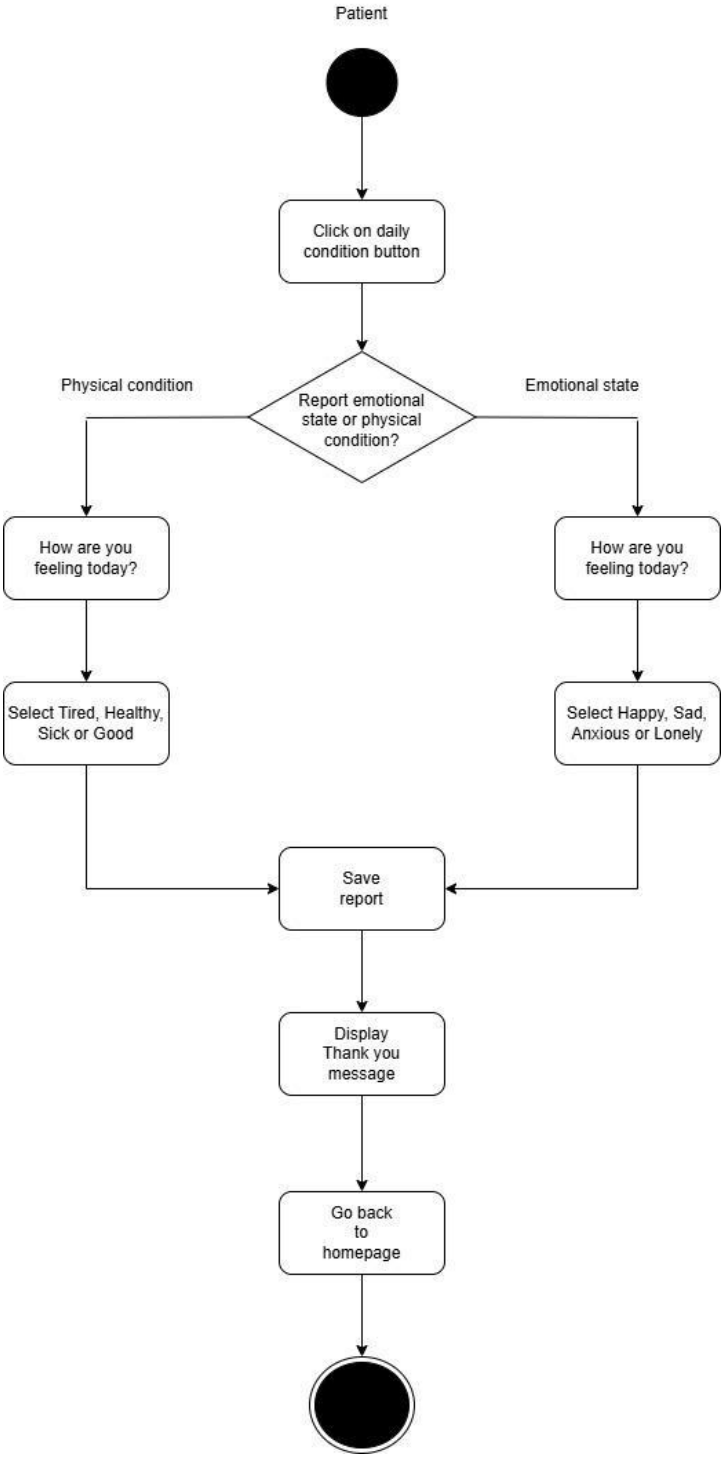
Background:
Given that I am a user
And I am on the homepage

Scenario: Select Physical Condition

When I press on the daily condition button
And I press report physical condition
Then I should see a message saying “How are you feeling physically today?”
And I should be able to select an option
And I should see a message saying “Thank you!”
And the system should save the reports in the daily condition
And I should be able to go back to homepage

Scenario: Select Emotional Condition

When I press on the daily condition button
And I press report emotional condition
Then I should see a message saying “How are you feeling emotionally today?”
And I should be able to select an option
And I should see a message saying “Thank you!”
And the system should save the reports in the daily condition
And I should be able to go back to homepage


Patient
Click on daily condition button
Report emotional state or physical condition?
Physical condition
Emotional state
How are you feeling today?
How are you feeling today?
Select Tired, Healthy, Sick or Good
Select Happy, Sad, Anxious or Lonely
Save report
Display Thank you message
Go back to homepage

2.2.12 Feature 12-View Daily Physical And Emotional Condition

View daily physical and emotional condition

- As a caregiver
- So that I can stay informed about my patient's condition
- I want to see their condition reports

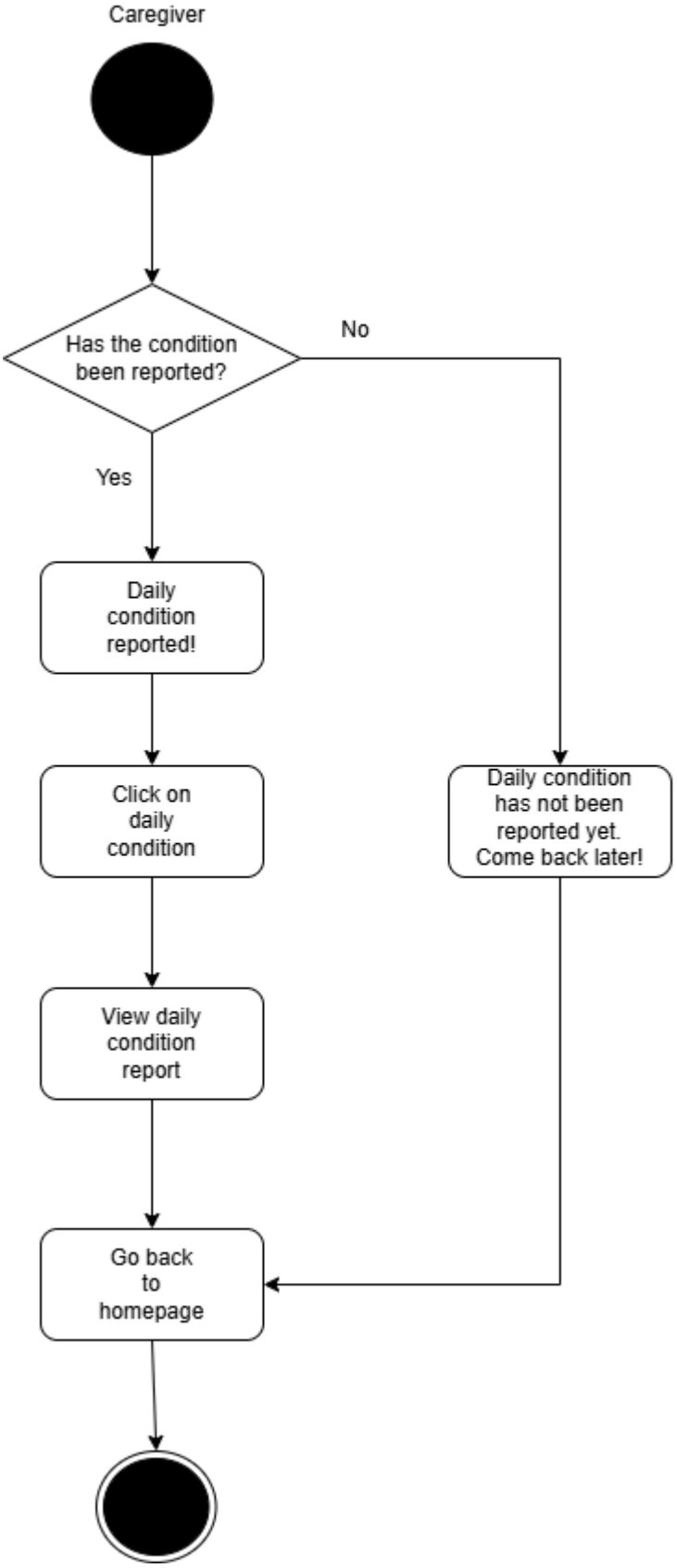
Background:
Given that I am a caregiver
And I am on the homepage

Scenario: Caregiver sees report that has been completed

When I press the daily condition
And the patient has already reported their condition
Then I should see a message saying "Daily Condition Completed!"
And I should be able to see the patient's physical condition and emotional state.
And I should be able to go back to homepage

Scenario: Caregiver sees report that has been uncompleted

When I press the daily condition
And the patient has not reported his condition yet
Then I should see a message saying "Daily Condition is not reported yet"
And I should be able to go back to homepage


Caregiver
Has the condition been reported?
No
Yes
Daily condition reported!
Click on daily condition
View daily condition report
Daily condition has not been reported yet. Come back later!
Go back to homepage

2.2.13 Feature 13-Wellness Reminders

<u>Wellness Reminders</u>

- As a patient
- So that I maintain my well-being
- I want to receive reminders to drink water and perform light exercise.

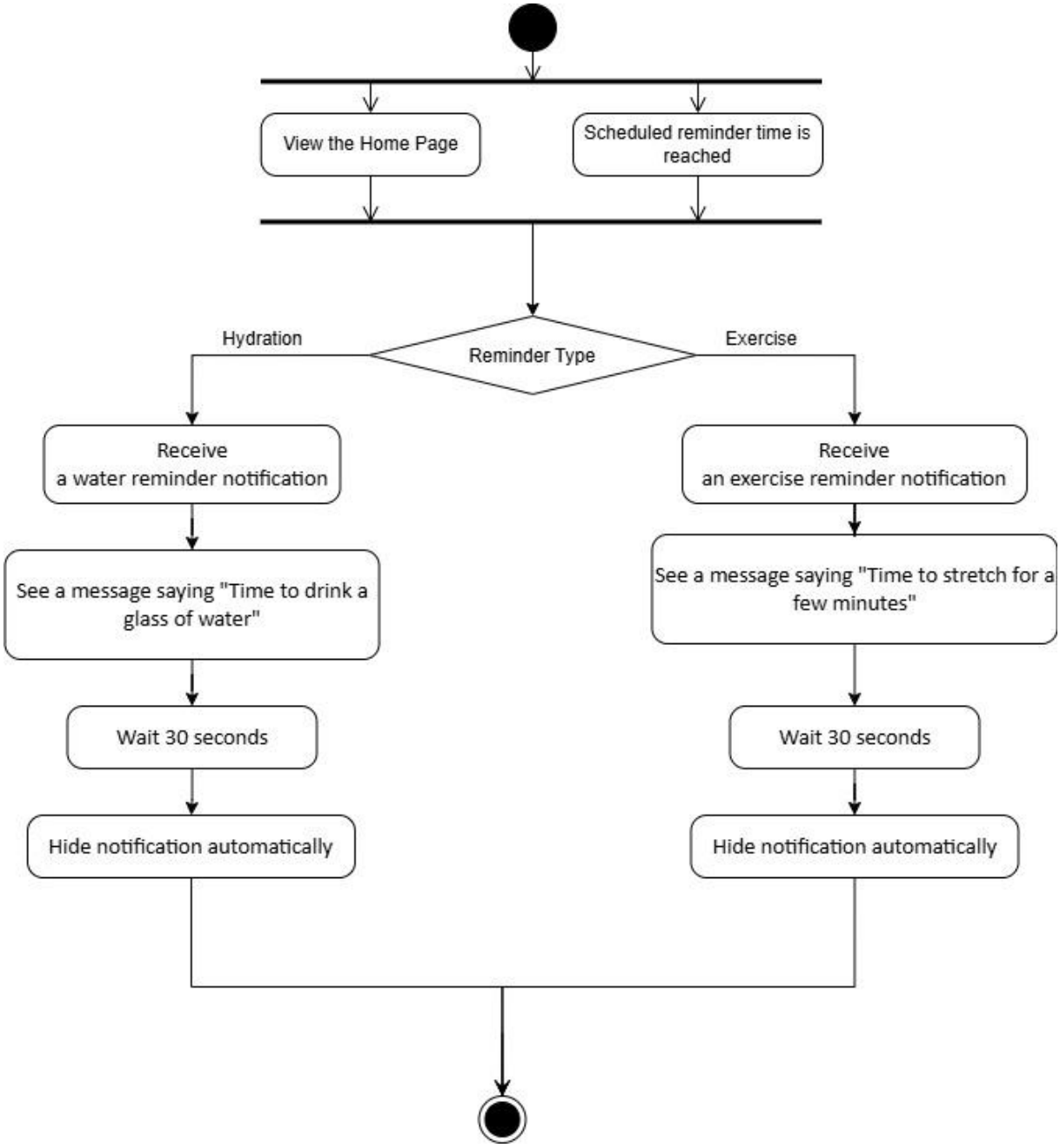
Background:
Given that I am the patient
And I am logged in
And I am on the homepage
And hydration reminder format is:
| hydration reminder | [12:00, 18:00, 21:00] |
And light exercise reminder format is:
| light exercise reminder | [12:15, 18:15, 21:15] |

Scenario: Patient receives a hydration reminder

When the hydration reminder time is reached
Then I should receive a water reminder notification
And I should see a water image on the screen
And I should see a message saying "Time to drink a glass of water"
And the notification should disappear after 30 seconds

Scenario: Patient receives an exercise reminder

When the exercise reminder time is reached
Then I should receive an exercise reminder notification
And I should see an exercise image on the screen
And I should see a message saying "Time to stretch for a few minutes"
And the notification message should disappear after 30 seconds


View the Home Page
Scheduled reminder time is reached
Hydration
Reminder Type
Exercise
Receive
a water reminder notification
Receive
an exercise reminder notification
See a message saying "Time to drink a glass of water"
See a message saying "Time to stretch for a few minutes"
Wait 30 seconds
Wait 30 seconds
Hide notification automatically
Hide notification automatically

2.2.14 Feature 14-Display Date and Time

Display Date and Time

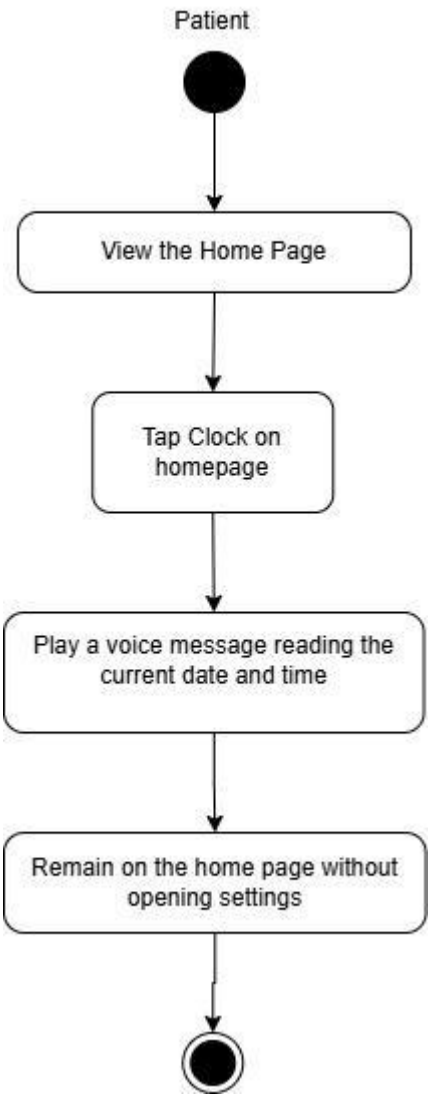
- As a patient
- So that I remain oriented and reduce confusion
- I want to clearly see the current date and time on the screen

Background:

Given that I am the patient
And I am logged in
And I am on the homepage
And the current time format is:
| time | [00–23]:[00–59] |
And the current date format is:
| date | [01–31]/[01–12]/[2026–2100] |
And the time should update automatically on the screen

Scenario: Patient taps the clock

When I press the clock on homepage
Then I should be able to hear the current date and time
And I should remain on the homepage without opening settings



2.2.15 Feature 15-Reminders On Locked Device

<u>Reminders on Locked Device</u>

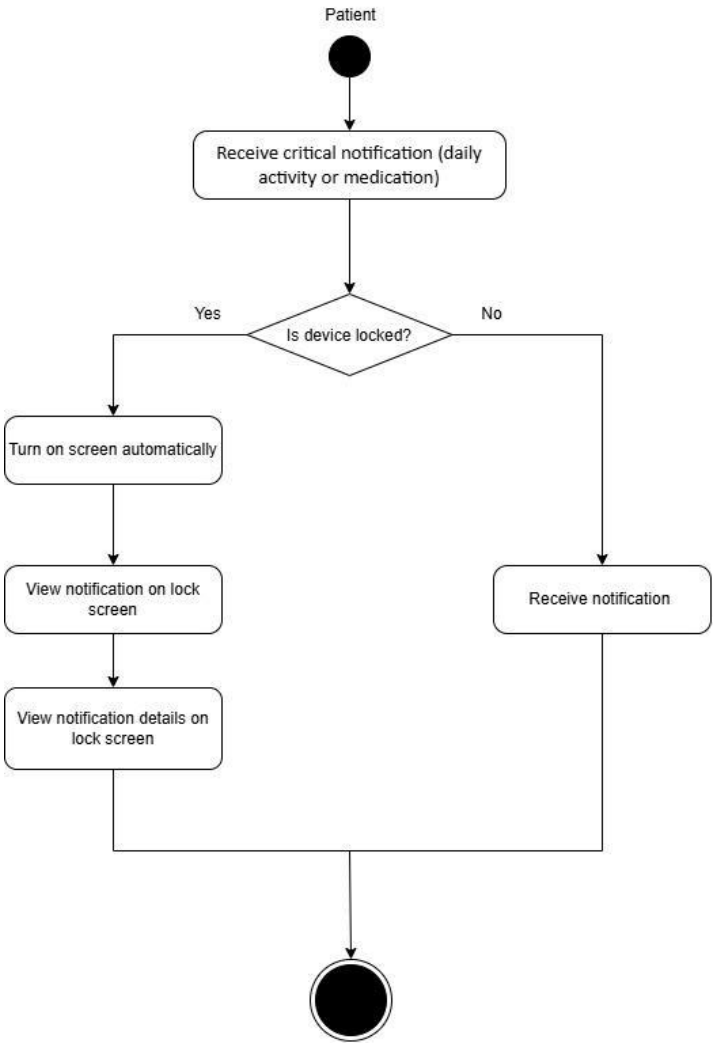
- As a patient
- So that I do not miss important reminders
- I want notifications to appear even when my device is locked

Background:

Given that the patient device is locked
And a medication notification or an activity notification is scheduled at a current time

Scenario: Critical reminder wakes the device

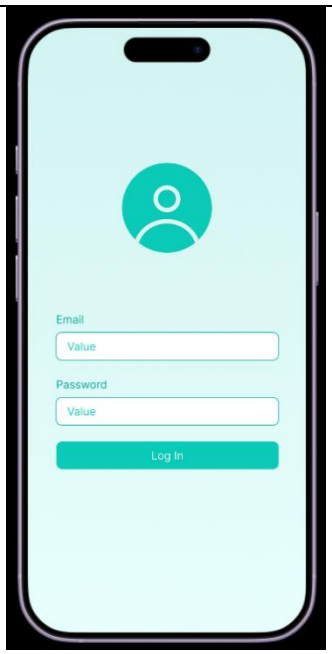
When I receive an activity or a medication notification
Then my device screen should turn on automatically
And I should see the notification details on my lock screen



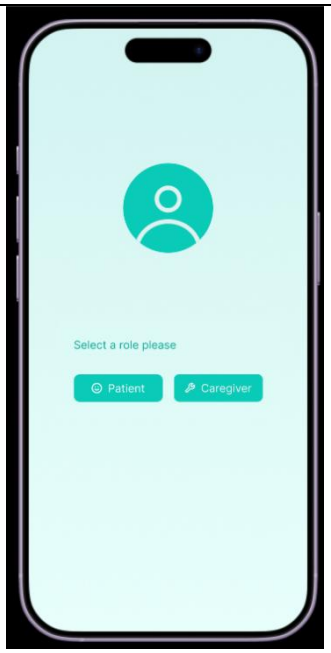
3 Indicative user interface mockups

3.1 Patient's view

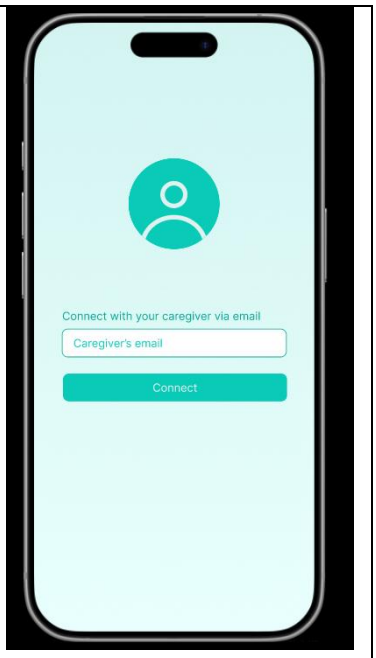
Opening screen



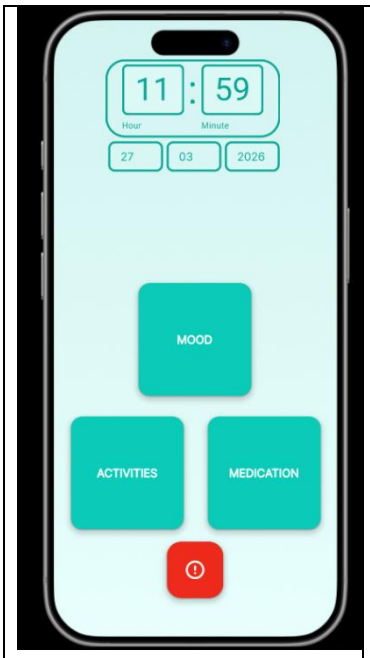
Log in page



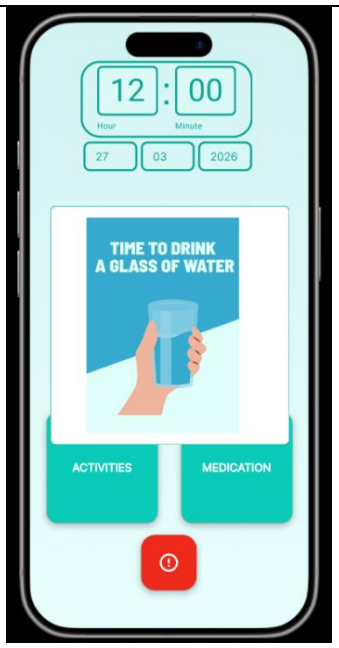
Select Role page



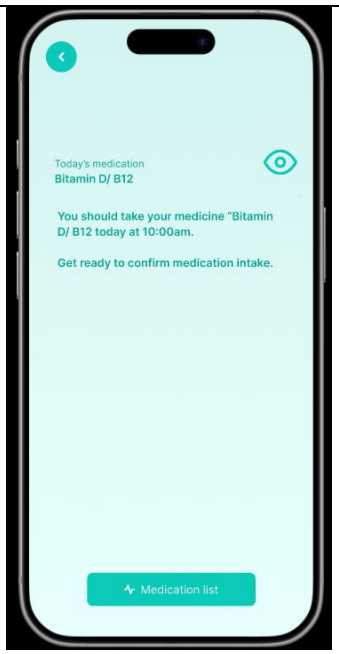
Connect with the caregiver



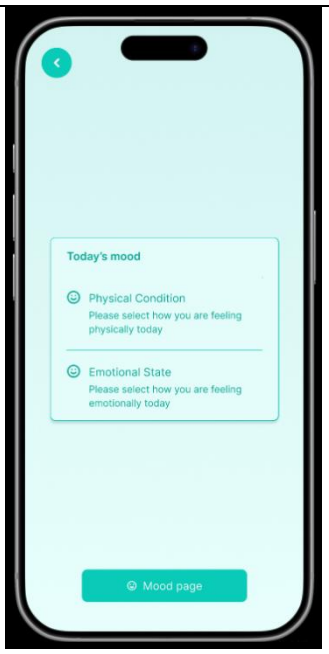
Patient's homepage



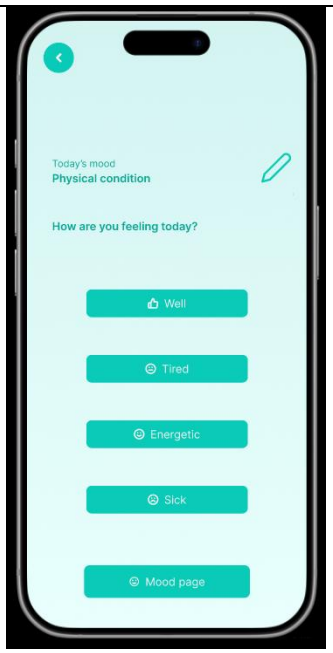
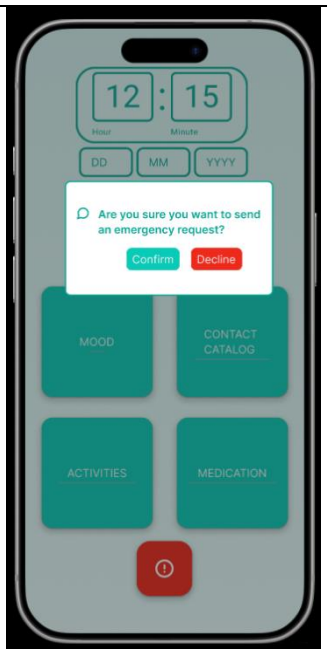
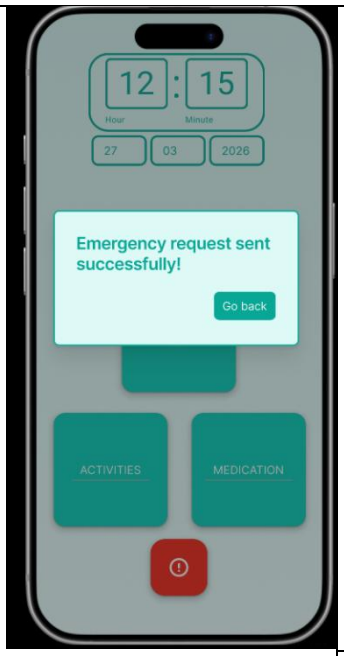
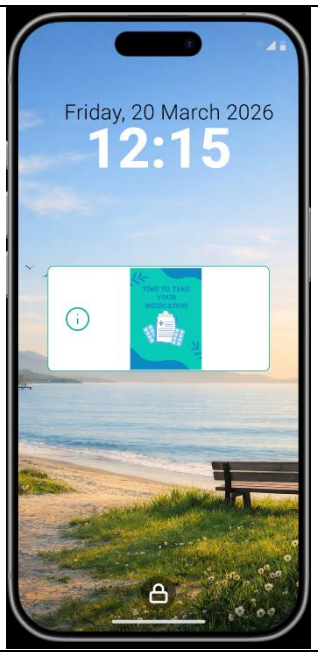
Patient receives wellness reminder on homepage



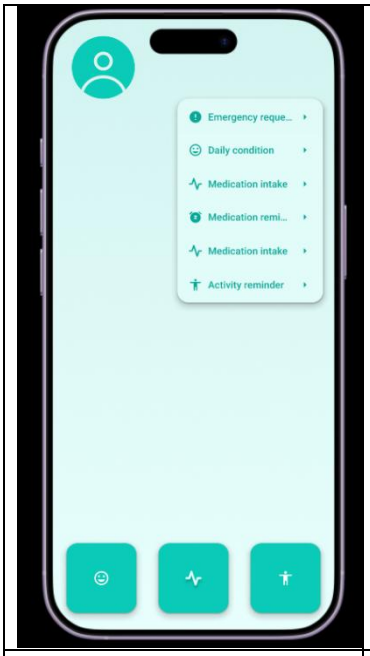
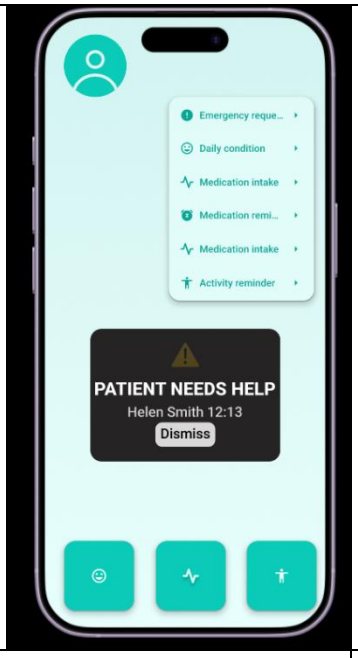
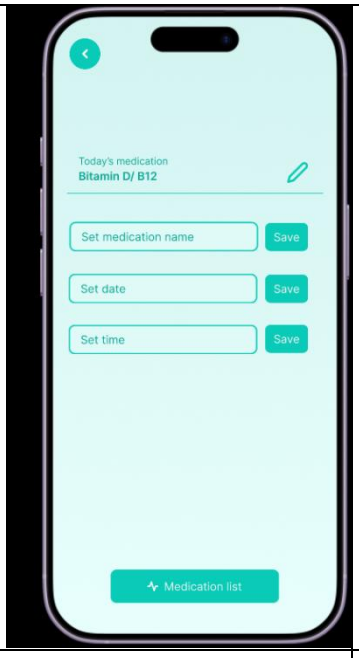
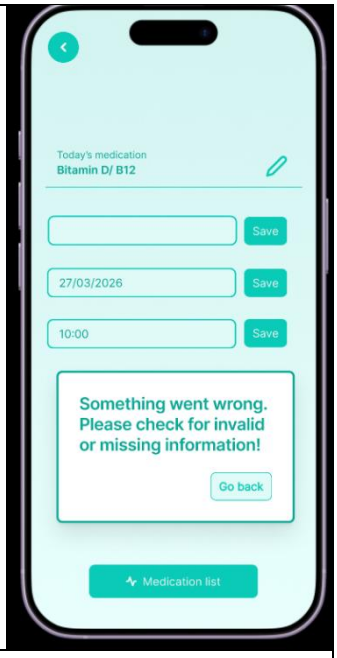
View an activity from the activity program page

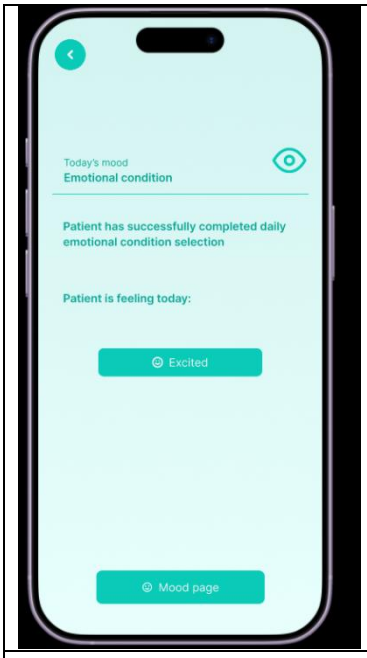
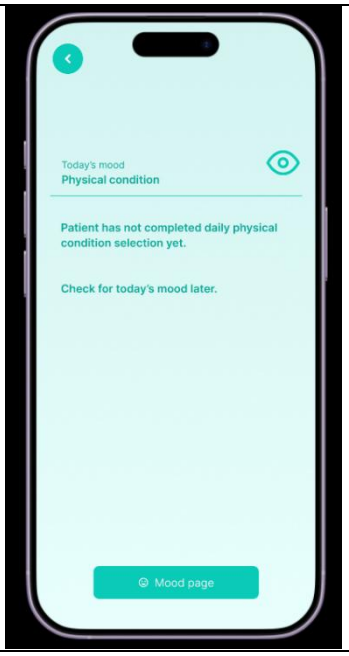
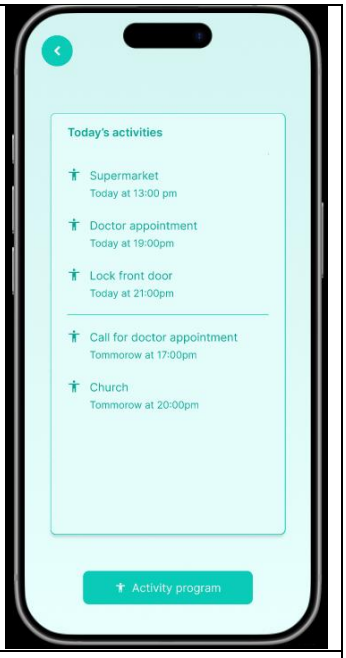


Daily condition page

			
Editing daily condition page	Receiving an emergency request page	Patient confirms he wants to sent emergency request	Patient's lockscreen with a medication notification

3.2 Caregiver's view

			
Caregiver's homepage	Caregiver receives an emergency request on homepage	Caregiver adds or edits medication	Caregiver enters invalid or empty info

		
Caregiver views completed daily condition	Caregiver views empty daily condition	View activity program page